Berkay Karaca

İzmir, Türkiye · github.com/Scryne · linkedin.com/in/berkay-karaca-3619a8419

SUMMARY

Software developer building end-to-end systems across multi-tenant SaaS, document AI, OSINT and industrial software. Pairs with AI coding agents and verifies every result with tests, evaluation sets and end-to-end runs on real data.

EDUCATION

Manisa Celal Bayar Üniversitesi — Manisa
Associate Degree in Computer Programming — 2025 – 2027 (expected)

- GPA: 3.72 / 4.00

PROJECTS *personal projects · built solo, pairing with AI coding agents*

TenderIQ — Citation-first AI analysis of public tender documents (multi-tenant SaaS) — Jul 2026 – Sep 2026
Python, FastAPI, Celery, PostgreSQL + pgvector, LangGraph, Next.js — github.com/Scryne/TenderIQ

- Built 5 LangGraph agents that extract requirements, risks, deadlines and eligibility gaps from Turkish tender PDFs; findings that cannot be tied to document, page and clause are rejected in code
- Implemented hybrid retrieval (BGE-M3 + BM25, RRF fusion, reranking) and a golden-set evaluation that fails CI when a prompt or model changes
- Isolated tenants with PostgreSQL row-level security; 415 unit + 183 integration tests; a sample tender yields 34 cited findings in ~5 min

FabrikaOS — Multi-tenant MES + ERP for manufacturing — Jun 2026 – Sep 2026
Python, FastAPI, PostgreSQL + TimescaleDB, MQTT, React, TypeScript — private repo

- Designed a schema-per-tenant architecture with 22 business modules (work orders, maintenance, quality, stock, HR) and live OEE ingestion from the shop floor over MQTT
- Added versioned scikit-learn models for predictive maintenance, anomaly detection and OEE forecasting
- Turned a CI red for 2 months green with hermetic tests and closed 10 high/critical advisories; 667 + 214 tests, 24 ADRs

AEGIS — KYB and due-diligence platform for AML compliance — Jul 2026 – Sep 2026
Python, FastAPI, PostgreSQL (RLS), Celery, OpenSanctions yente, Next.js — private repo

- Built sanctions/PEP screening on a self-hosted OpenSanctions engine with Turkish name normalization, analyst decisions with rationale and a hash-chained audit trail
- Stored national ID numbers encrypted with blind indexing and sealed case reports with SHA-256
- Found and fixed extraction linking a second director to the first one's ID number in the first end-to-end run; 259 unit + 79 integration + 36 recall tests

GhostBuilding — Censorship and change detection in satellite imagery (OSINT) — Apr 2026 – Sep 2026
Python, FastAPI, PostGIS, OpenCV, Celery, Next.js, MapLibre — github.com/Scryne/GhostBuilding

- Developed an engine that compares Esri, MapTiler and IGN tiles per location with SSIM, Laplacian variance and FFT to flag blurred, mosaicked or mismatched imagery
- Measured uncalibrated thresholds flagging 39 of 40 ordinary locations; after calibrating on a 60-location baseline, a 27-site scan flagged 3 military sites and 0 control sites

Wartrack — Regional news monitoring with an explainable threat level — Feb 2026 – Sep 2026
TypeScript, Node.js, Express, Socket.IO, SQLite, React, Leaflet, Ollama — github.com/Scryne/Wartrack

- Mapped 15 news feeds to places with a gazetteer; the 1–5 threat level and its reasons are computed in code, a local LLM only writes Turkish summaries
- Fixed a fetch stall on slow DNS (0/15 → 15/15 feeds) and closed a DNS-rebinding gap by validating addresses at connect time

Also: MythAtlas (mythology atlas with verified sources) · Kokpit (Electron app for AI-assisted development) · PolitikDesk (political intelligence desk with deterministic evidence rules)

TECHNICAL SKILLS

Languages	Python, TypeScript, JavaScript, SQL
Backend	FastAPI, Celery, SQLAlchemy, Alembic, Node.js, Express, Socket.IO
Data	PostgreSQL (RLS, pgvector, PostGIS, TimescaleDB), Redis, SQLite, MinIO, MQTT
Frontend	React, Next.js, Tailwind CSS, Vite, MapLibre, Leaflet, ECharts
AI / ML	LangGraph, RAG (hybrid retrieval), evaluation sets, scikit-learn, OpenCV, Ollama, Claude Code
Testing & DevOps	pytest, Vitest, Playwright, testcontainers, Docker, GitHub Actions, mypy, Ruff, gitleaks
Spoken	Turkish (native), English (B1)